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% The purpose of this script is to work with MODIS data

Load meshes

As built by functions in the ALOC8/Data/Omega folder

```
load('C:/Users/Alex/Documents/MATLAB/ALOC8/Datasets/Omega/OUTPUT/  
meshes.mat'); %Consult structure in workspace for formatting
```

MODIS MCD14ML

```
%years = { '/2010','/2011','/2012','/2013','/2014','/2015' };  
%MODISroot = 'C:/Users/Alex/Documents/MATLAB/ALOC8/Datasets/Ongoing  
Fire/MODIS';  
%for yy = 3:3:numel(years)  
%   FileNames = gunzip([ MODISroot '/GZ' years{yy} '/*.gz' ],...  
%       [ 'C:/Users/Alex/Documents/MATLAB/ALOC8/Datasets/Ongoing  
Fire/MODIS/ASC' years{yy} ]);  
%end  
%delete(FileNames{:}); %Clear directory
```

MODIS RSAC routines

```
filelist = dir('C:/Users/Alex/Documents/MATLAB/ALOC8/Datasets/Ongoing  
Fire/MODIS/TXT/2012'); %List files in the directory  
lon = []; lat = []; FRP = []; tt = []; %Initialize variables to load  
for ff = 3:numel(filelist) %For each file skipping directories  
    dat = readtable(filelist(ff).name, 'Delimiter', '\t'); %Read file  
    into table  
    dat(dat.Confidence < 80.0,:) = []; %Only save FRP Confidence  
    80-100% (see Hereborn et al 2014)  
    lon = [ lon;dat.Longitude ]; %Parse longitude  
    lat = [ lat;dat.Latitude ]; %Parse latitude  
    FRP = [ FRP;dat.FRP ]; %Parse fire radiative power  
    %Parse time stamp  
    tt =  
    [ tt;datetime(strcat(dat.DetectionDate,dat.DetectionTimeUTC), 'InputFormat', 'dd  
MMM yyyyHH:mm') ];  
end  
% Build some test plots  
T0 = datenum(2012,7,20); TF = datenum(2012,7,29); TSTEP =  
    0.5; %Initial time, final time, time step  
tComb = (T0:TSTEP:TF)'; %Forming a comb/partition of the time interval
```

```

tt = datenum(tt); %Convert to datenum format
[numPts,~,bins] = histcounts(tt,tComb); %Bin times by times according
to tComb
xx = mesh(13).grid(mesh(13).in,1); %x-coordinates
yy = mesh(13).grid(mesh(13).in,2); %y-coordinates
for nn = 1:(numel(tComb)-1) %For each time step
    disp(['Time step ' num2str(nn) ' @ '
num2str(tComb(nn)) '...']); %Report progress to user
    tf = (bins == nn*ones('like',bins)); %Boolean indicator for
current time interval
    lon2plot = lon(tf); %Peel longitudes
    lat2plot = lat(tf); %Peel latitudes
    FRP2plot = FRP(tf); %Peel FRP values
    if ~isempty(lon2plot)
        % Cluster the coordinates to try to capture single fire
averages
        MaxClusterSep = inf; %Initialize cluster separation
        capK = 1; %Number of clusters
        while MaxClusterSep > 0.83 %Threshold in degrees separation
            capK = capK + 1; %Bump up number of clusters
            [idx,lonlat2interp] = kmeans([lon2plot
lat2plot],capK); %k-means subroutine
            MaxClusterSep = min(pdist(lonlat2interp)); %Compute the
maximum of cluster centroid spacings
        end
        % Look into the distribution of FRP measurements
        LARGE_FIRE = 18; %Initialize large fire size
        lf = 0; %Initialize large fire count
        FRP2interp = NaN*ones(capK,1); %Initialize FRP means or
whatever
        for kk = 1:capK %For each cluster
            if sum(idx == kk) > LARGE_FIRE %If there is some binning
worth doing (this is a definition of large fire?)
                lf = lf + 1; %Iterate large fire count
                if nn == 2
                    figure; cla; h = histogram(FRP(idx == kk));
title([num2str(sum(idx == kk)) ' Measurements of "Large" Fire at ('
num2str(lon2plot(kk)) ', ' num2str(lat2plot(kk)) ')']);
                    view(0,90); xlim(h.BinLimits); ylim([0
max(h.Values)]); pause(0.5);
                end
                % TODO: Fit exponential distribution (TRY WEIBULL)
            end
            % Take average value for each cluster at centroid (this
could be any statistic)
            FRP2interp(kk) = max(FRP2plot(idx == kk));
            %FRP2interp(kk) = mean(FRP2plot(idx == kk));
        end
        disp([' -> Found ' num2str(lf) ' LARGE_FIRES.']);
        % Annex corner zeros for better interpolation
        ptA = [ min(mesh(13).outline(:,1))
min(mesh(13).outline(:,2)) ];
        ptB = [ max(mesh(13).outline(:,1))
min(mesh(13).outline(:,2)) ];

```

```

        ptC = [ max(mesh(13).outline(:,1))
max(mesh(13).outline(:,2)) ];
        ptD = [ min(mesh(13).outline(:,1))
max(mesh(13).outline(:,2)) ];
        lonlat2interp = [ lonlat2interp;ptA;ptB;ptC;ptD ]; %Append to
coordinate vector
        FRP2interp = [ FRP2interp;zeros(4,1) ]; %Append to values to
be interpolated
        % MATLAB's interpolation routine (natural for C^2 functions)
        F =
scatteredInterpolant(lonlat2interp(:,1),lonlat2interp(:,2),FRP2interp,'natural');
        Z = F(xx,yy);
        % griddata provides a different/slower approach, the 'v4'
setting is awesome
        %[xq,yq] = meshgrid(xx,yy);
        %Z =
griddata(lonlat2interp(:,1),lonlat2interp(:,2),FRP2interp,xq,yq,'cubic');
        %cla; surf(xq,yq,Z,'EdgeAlpha',0.25); hold on; pause
        % Report result to user
        disp(['      (Number of clusters,MaxClusterSep) =      ('
num2str(capK) ', ' num2str(MaxClusterSep) ').']); %Print diagnostic
    else
        Z = zeros(numel(xx),1);
        idx = [1 1 1];
    end
    figure; cla; hold on;
    scatter3(lon2plot,lat2plot,FRP2plot,25,idx,'filled'); %Plot clusters
in different colors
    scatter3(xx,yy,Z,'r. '); view(-35,38); grid('on'); xlim([-140
-60]); ylim([25 50]);
    xlabel('Longitude'); ylabel('Latitude'); zlabel('FRP (MW)');
    title(['DOY = ' num2str(tComb(nn) - datenum(2012,1,1))]);
    pause(0.5);
        %SLIM = 0.105; %Parameter for "globality" or "demand
loudness", larger values give Dirac-delta type bumps
        % Could we vary the urgency here by proximity to WUI, fire
size, etc...?
        %Z =
expBumpGenr8er(coords(:,1),coords(:,2),xq,yq,SLIM*ones(numel(coords(:,1)),1),2);
        %surf(xq,yq,z); pause
        %scatter(coords(:,1),coords(:,2)); view(0,90); pause(0.25);
end

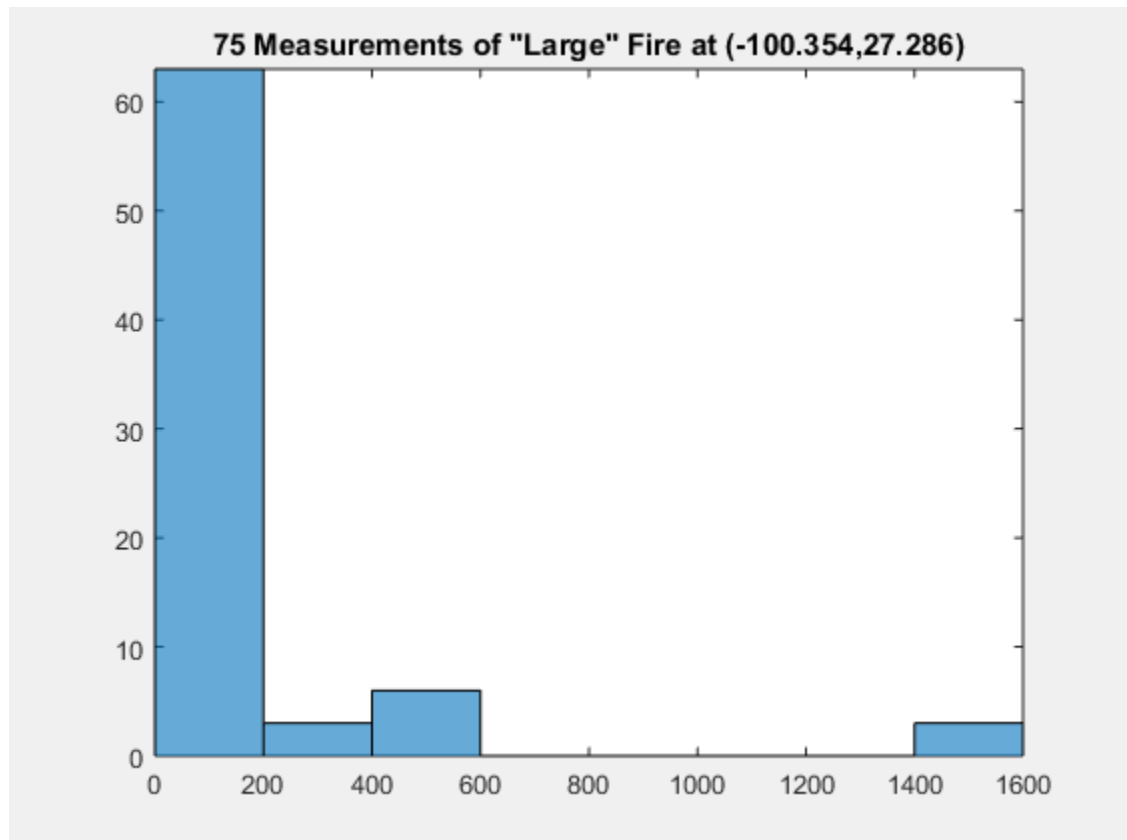
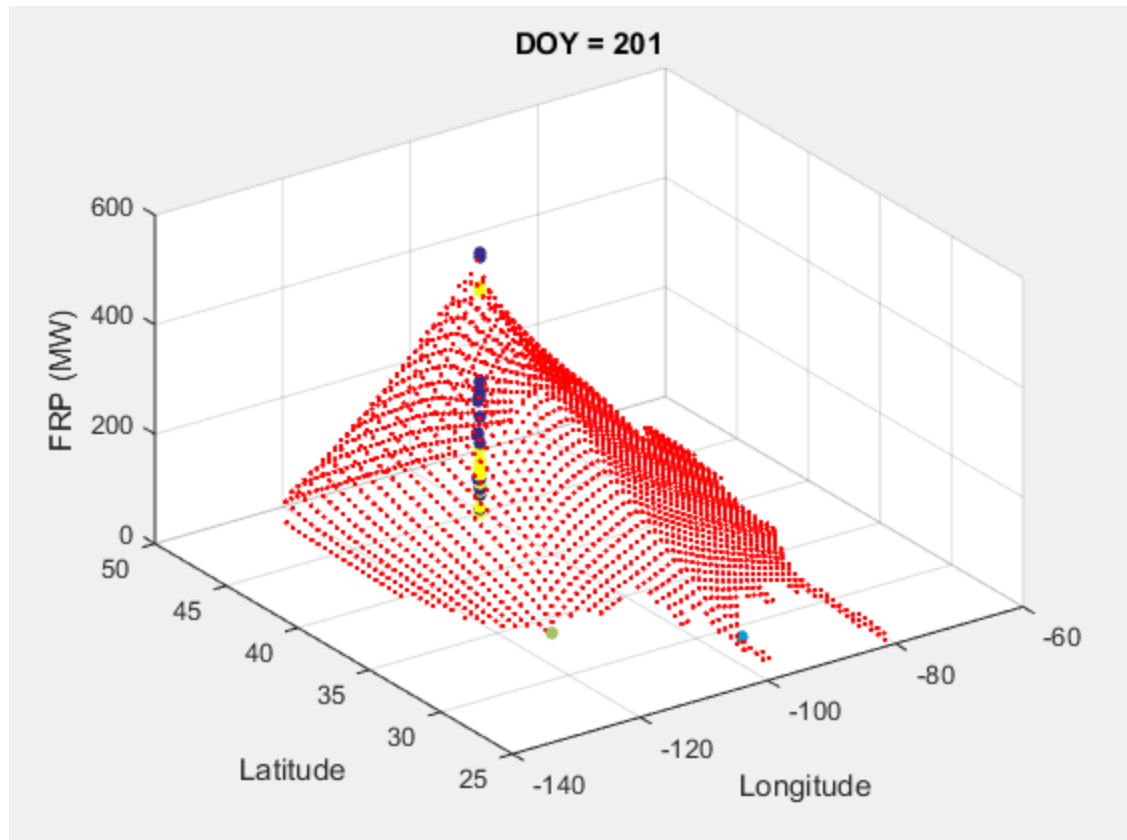
Time step 1 @ 735070...
    -> Found 2 LARGE_FIRES.
        (Number of clusters,MaxClusterSep) =      (4,0.034619).
Time step 2 @ 735070.5...
    -> Found 6 LARGE_FIRES.
        (Number of clusters,MaxClusterSep) =      (20,0.80716).
Time step 3 @ 735071...
    -> Found 0 LARGE_FIRES.
Warning: Duplicate data points have been detected and removed -
corresponding
values have been averaged.

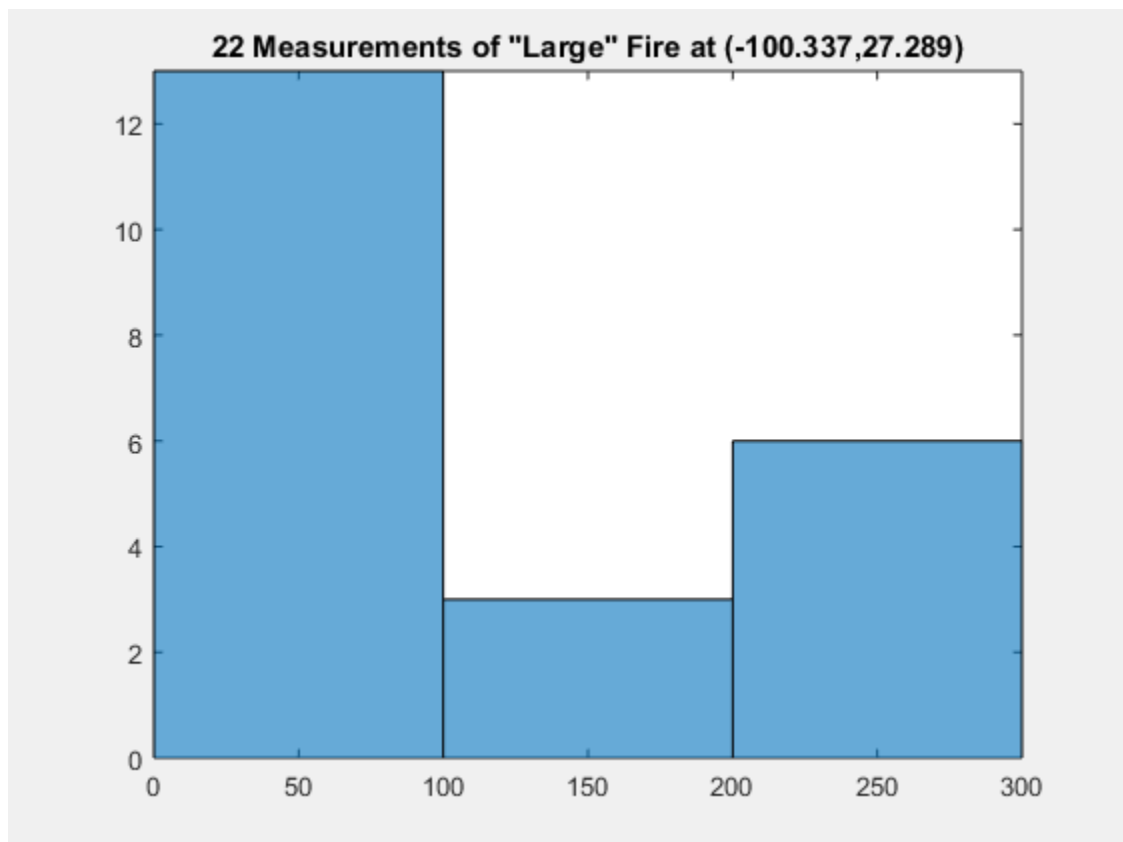
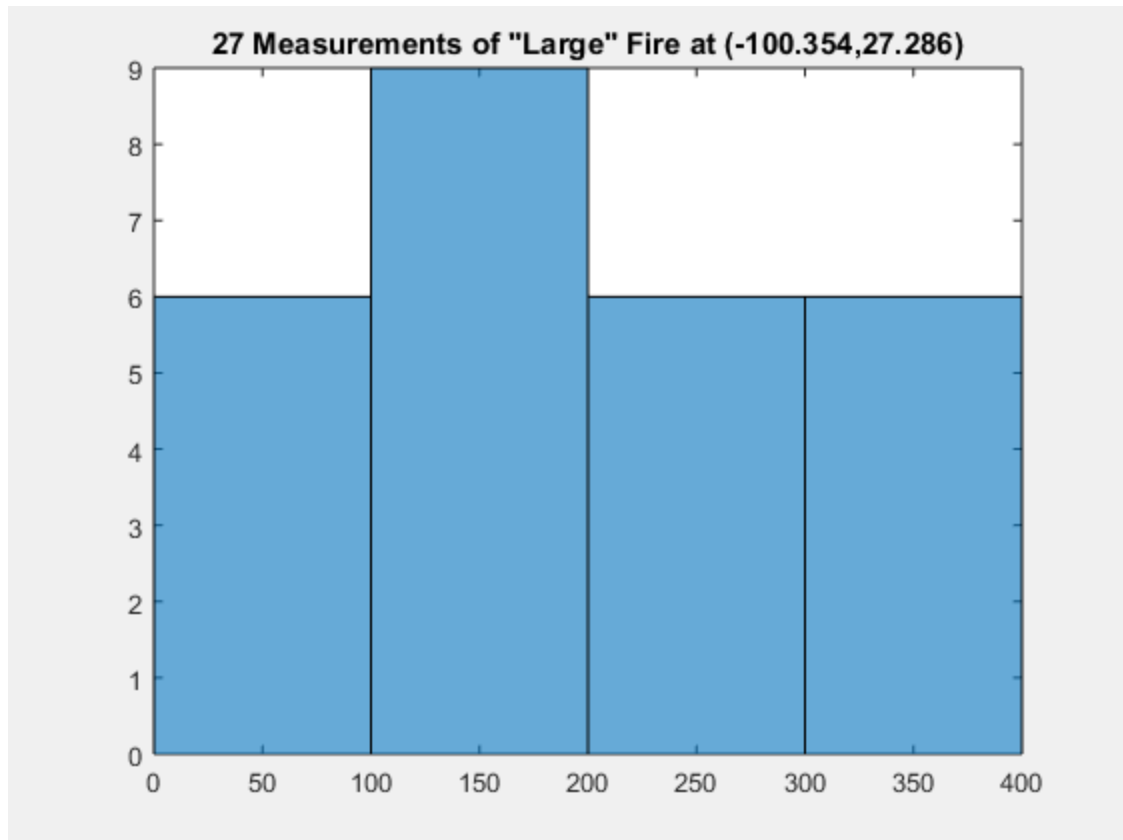
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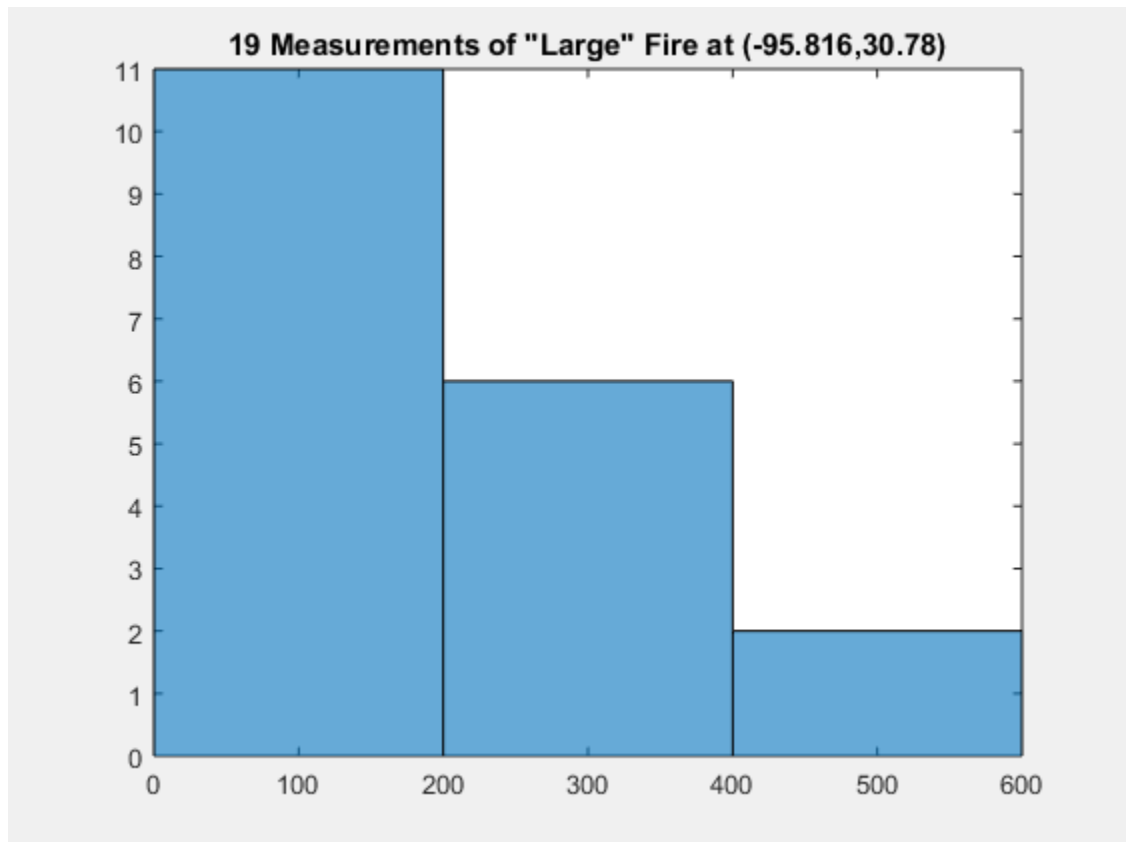
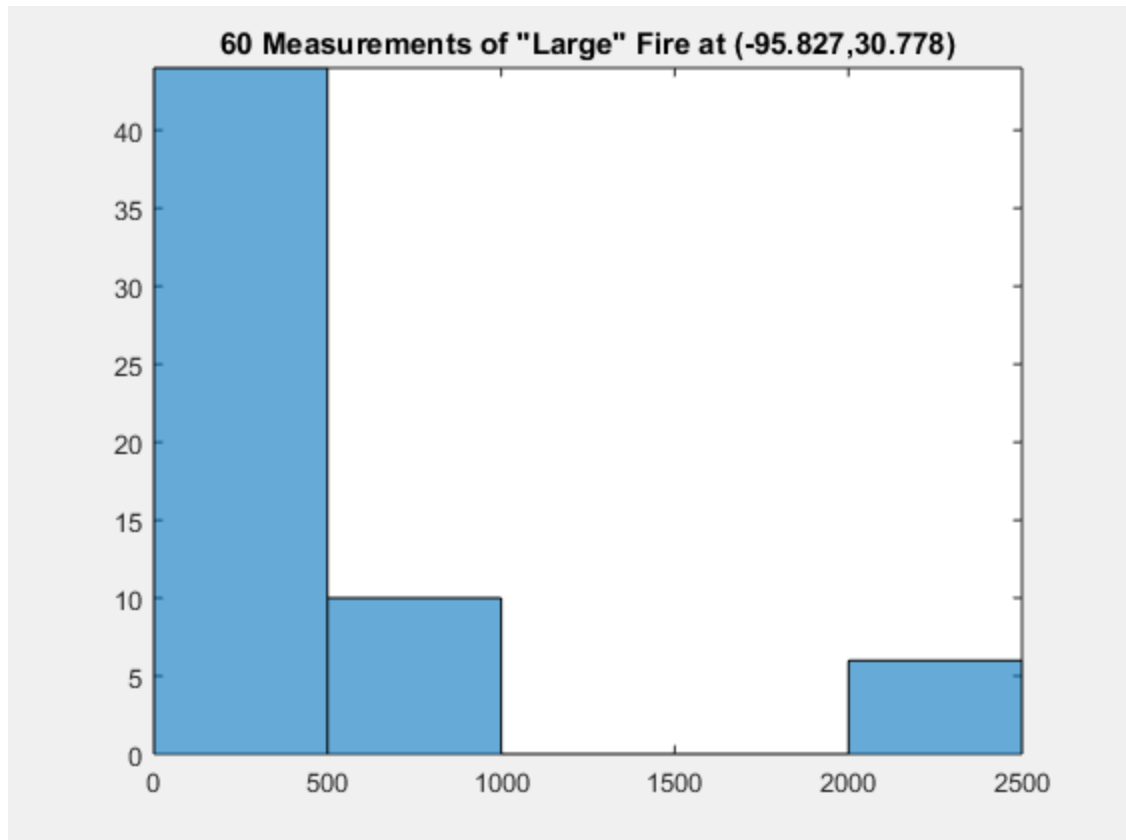
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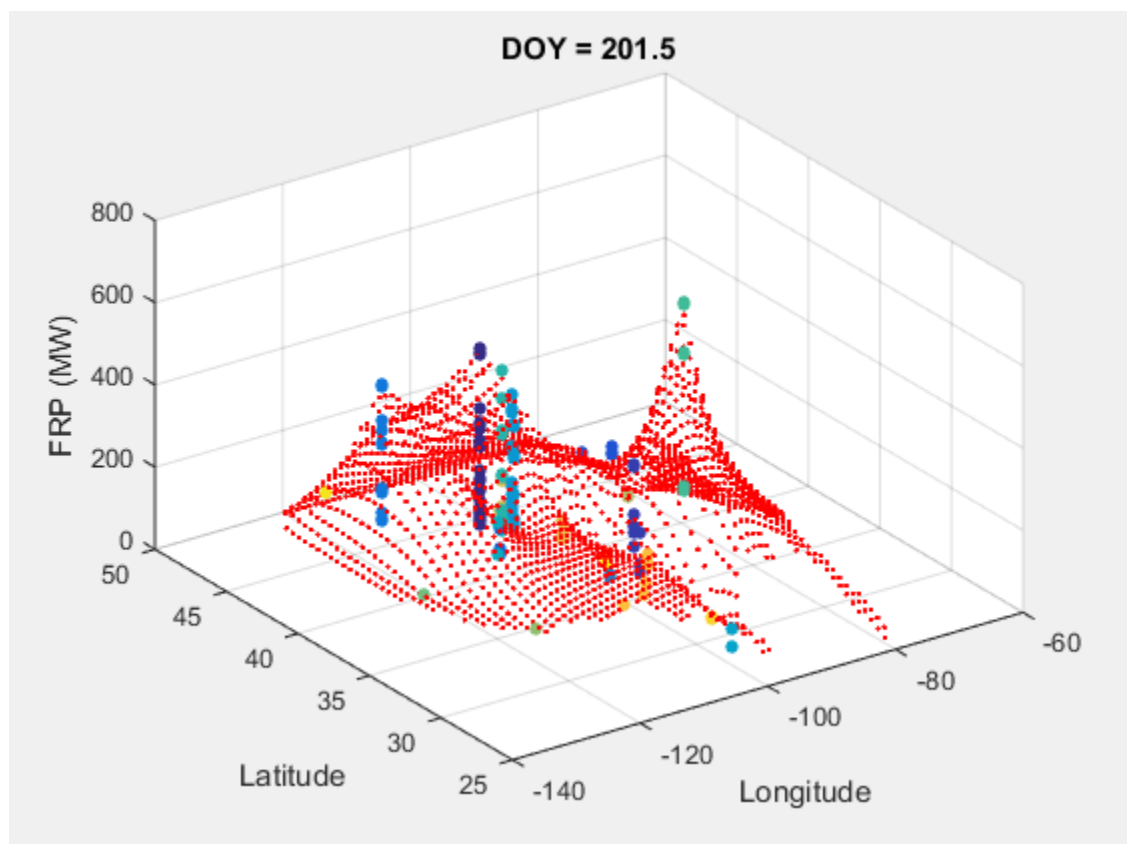
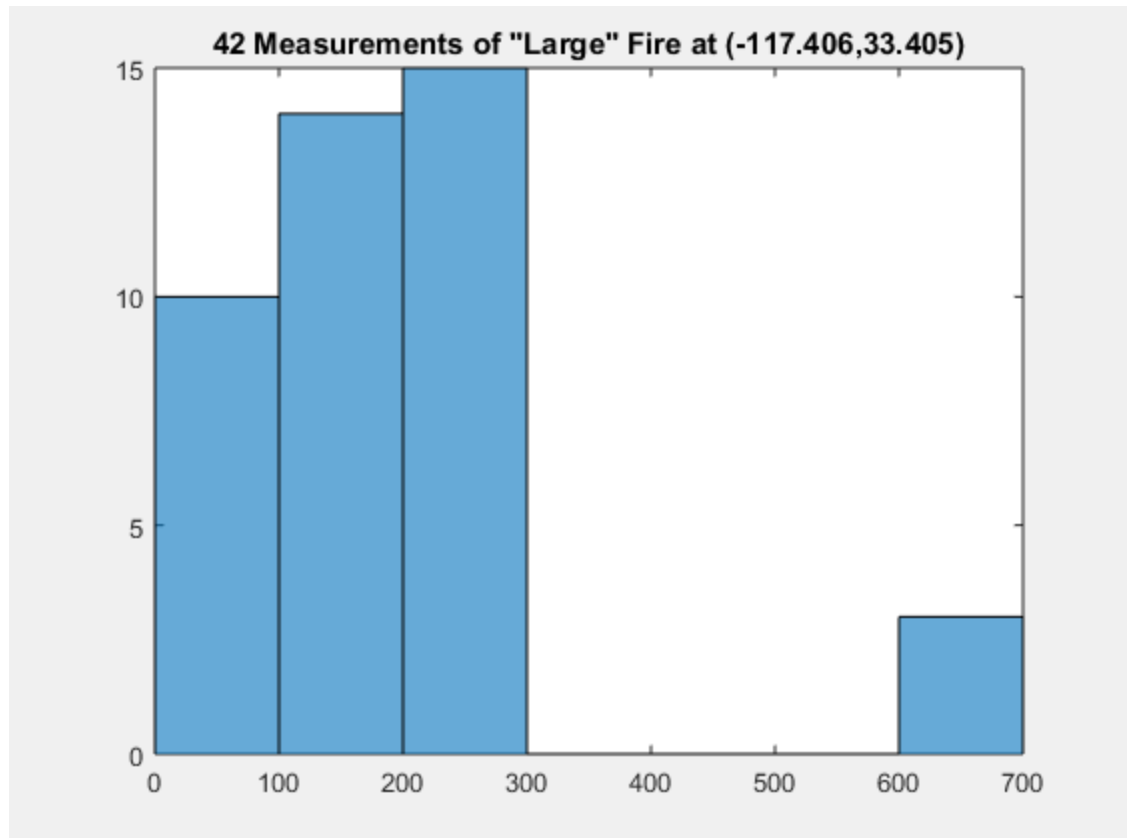
      (Number of clusters,MaxClusterSep) =      (7,0).
Time step 4 @ 735071.5...
  -> Found 2 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (19,0.79053).
Time step 5 @ 735072...
  -> Found 0 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (4,0.016935).
Time step 6 @ 735072.5...
  -> Found 6 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (21,0.41409).
Time step 7 @ 735073...
  -> Found 0 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (3,0.010349).
Time step 8 @ 735073.5...
  -> Found 3 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (17,0.8005).
Time step 9 @ 735074...
  -> Found 0 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (6,0.017208).
Time step 10 @ 735074.5...
  -> Found 1 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (15,0.72343).
Time step 11 @ 735075...
  -> Found 0 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (4,0.25506).
Time step 12 @ 735075.5...
  -> Found 4 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (14,0.68696).
Time step 13 @ 735076...
  -> Found 2 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (6,0.22336).
Time step 14 @ 735076.5...
  -> Found 2 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (18,0.78164).
Time step 15 @ 735077...
  -> Found 0 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (2,0.029347).
Time step 16 @ 735077.5...
  -> Found 0 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (18,0.53087).
Time step 17 @ 735078...
  -> Found 3 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (6,0.01688).
Time step 18 @ 735078.5...
  -> Found 4 LARGE_FIRES.
      (Number of clusters,MaxClusterSep) =      (19,0.067552).

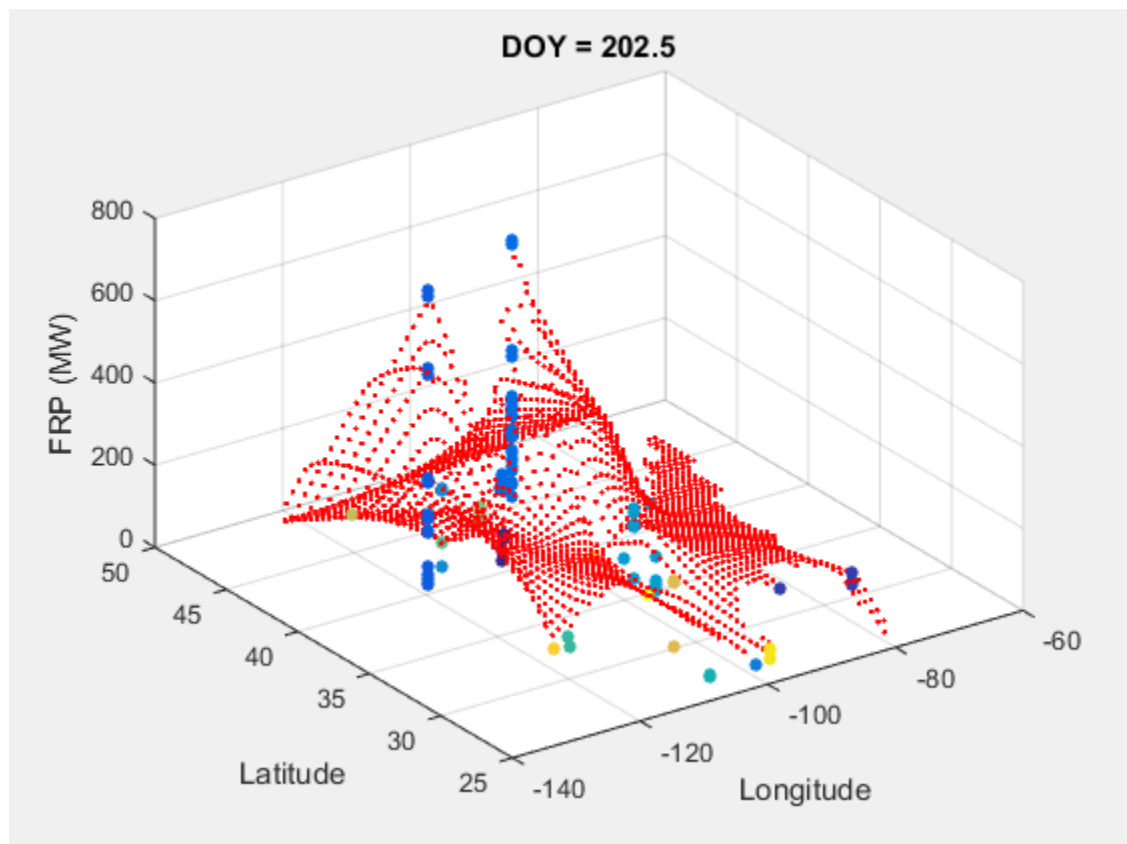
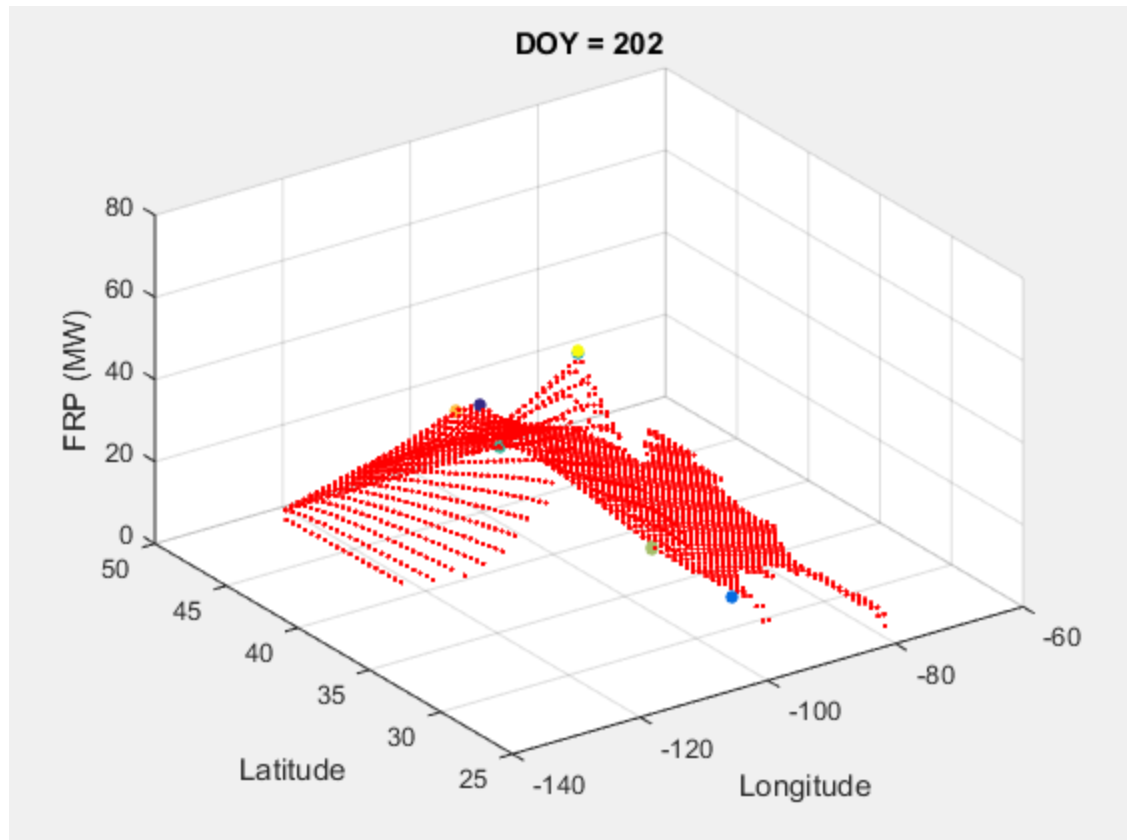
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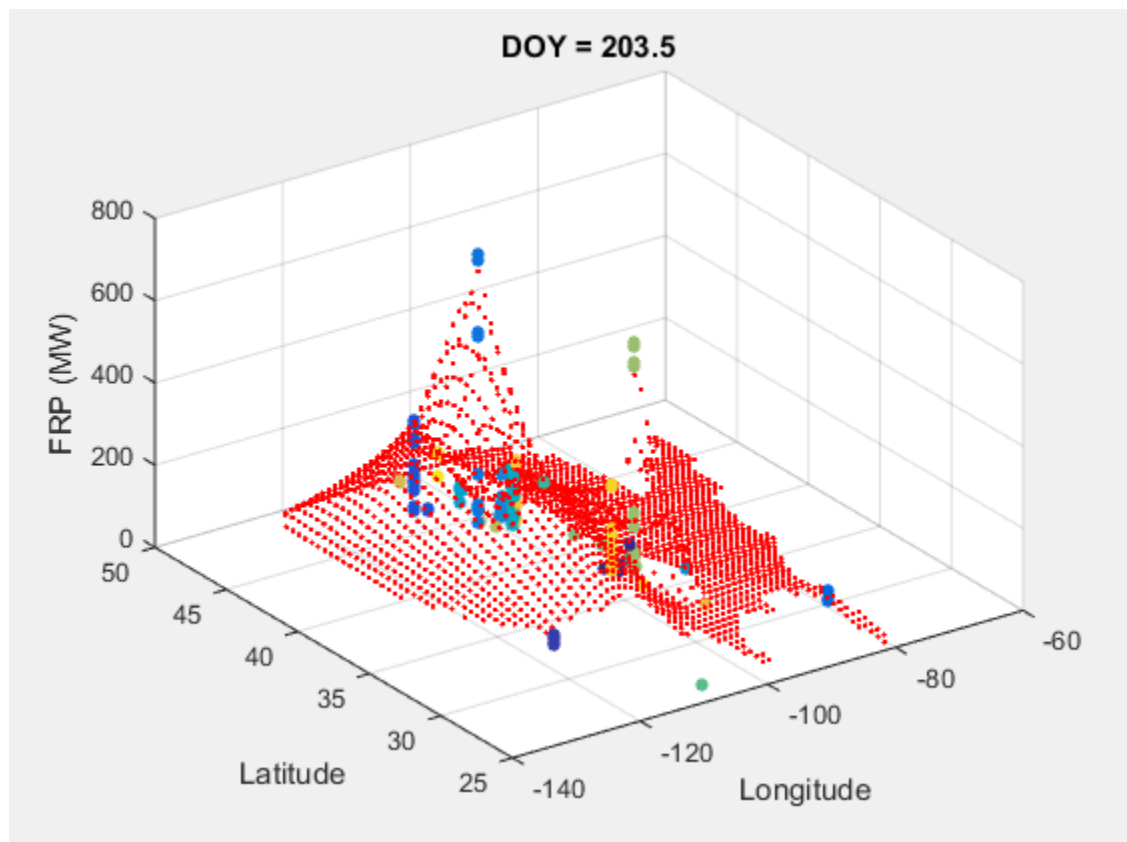
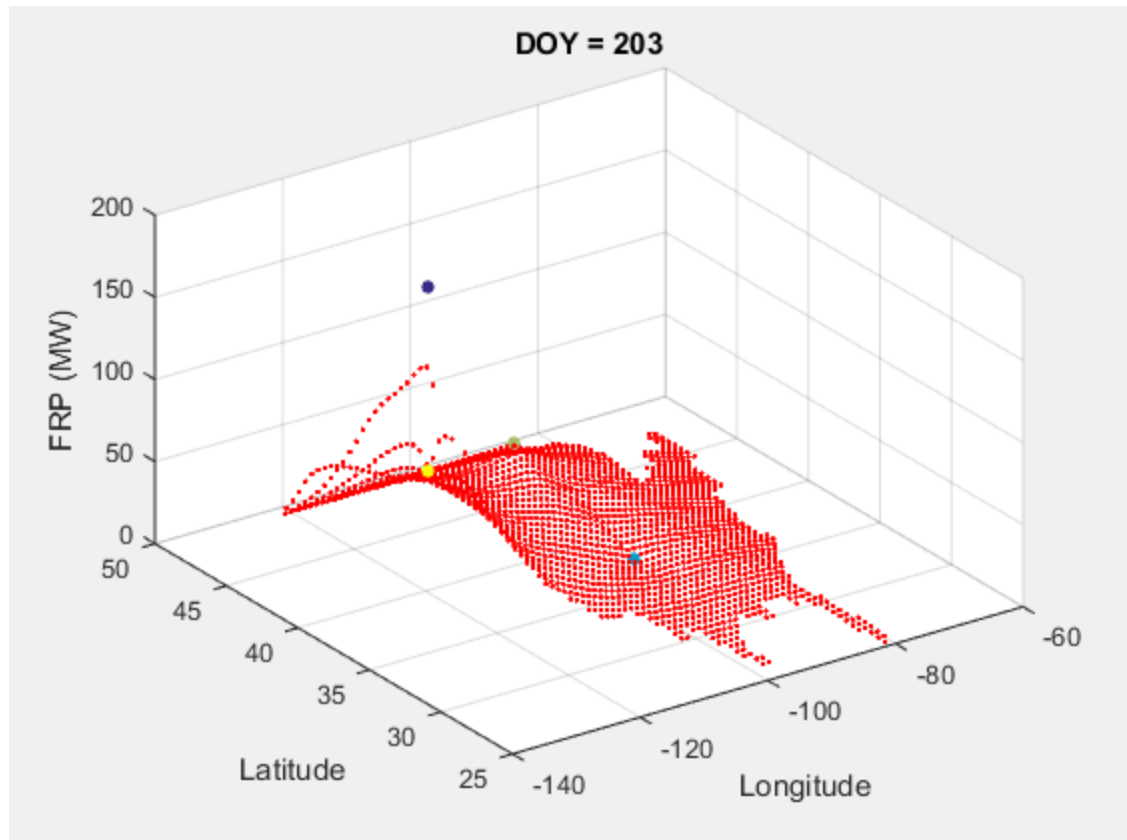


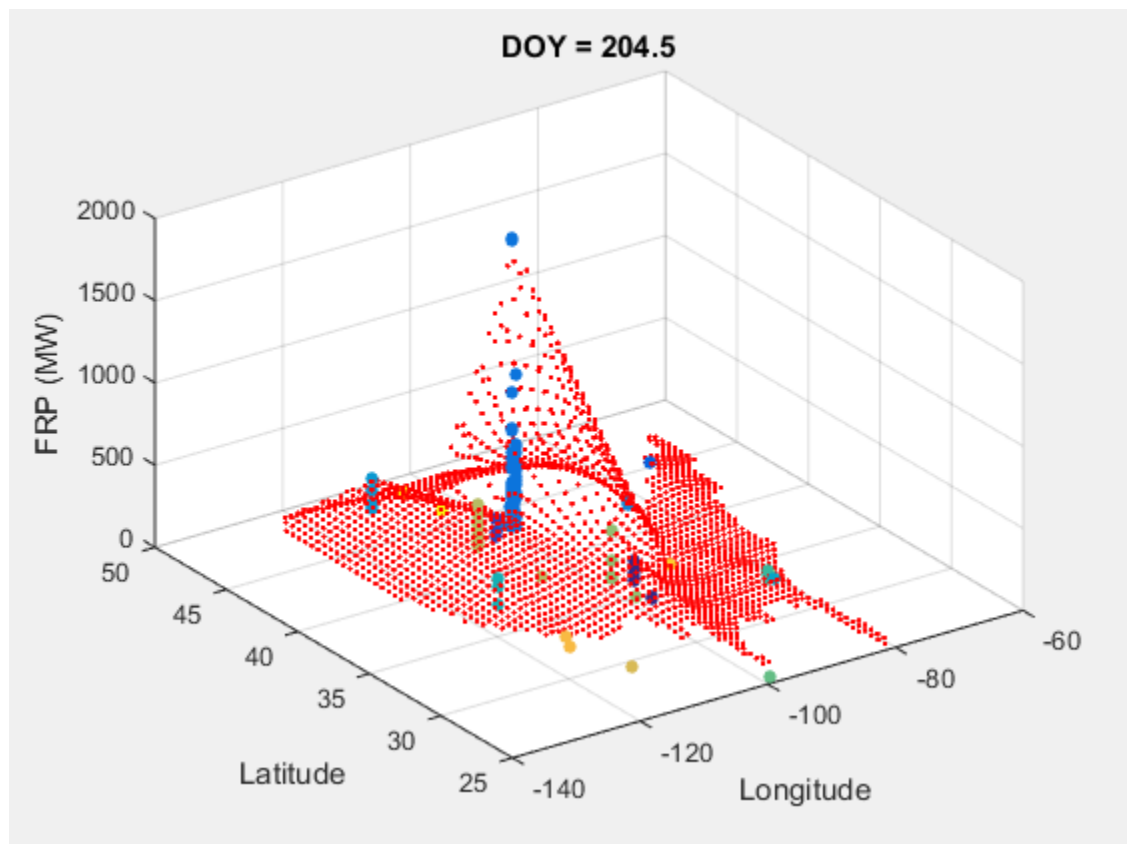
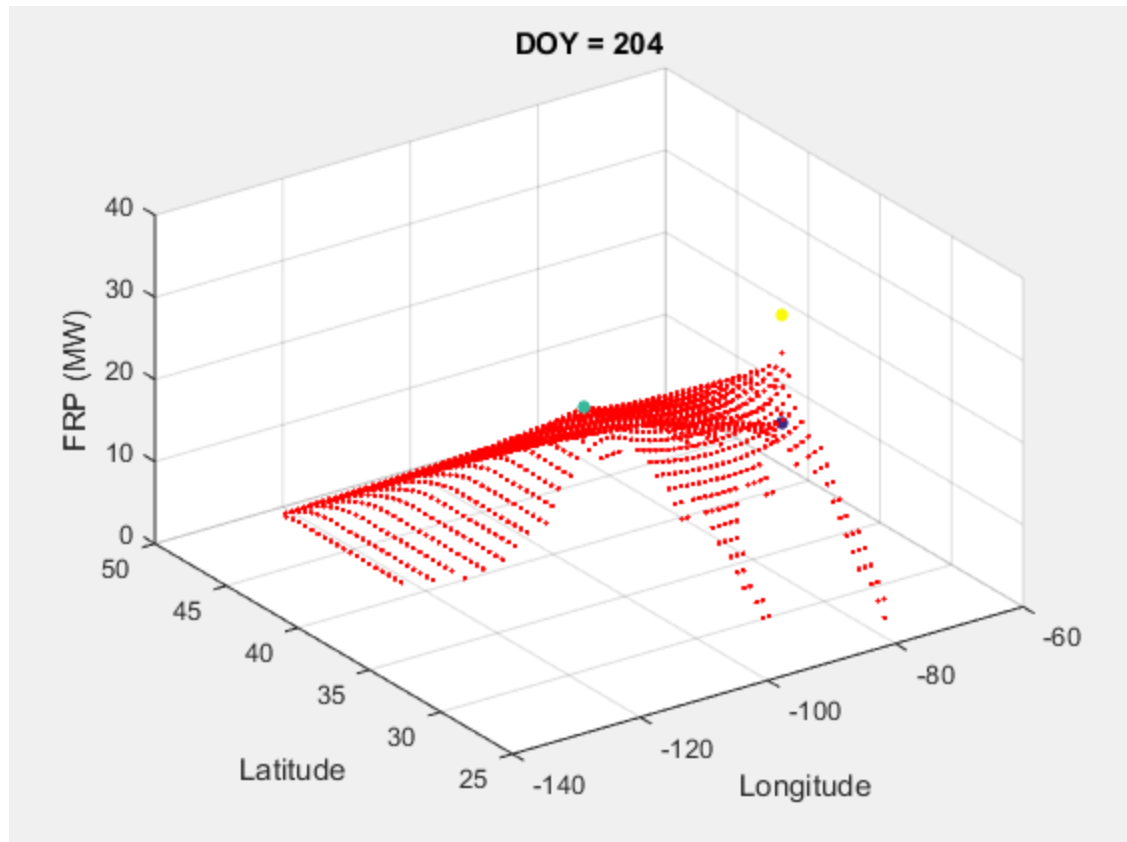


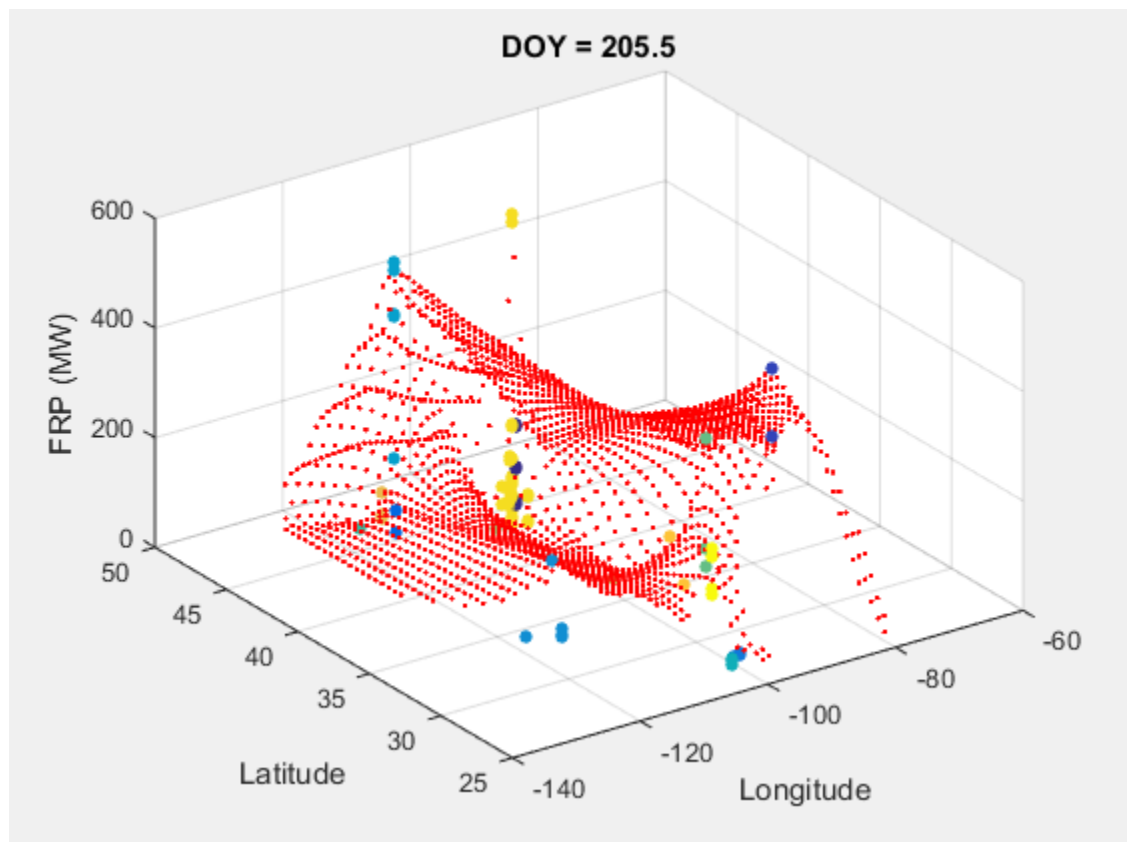
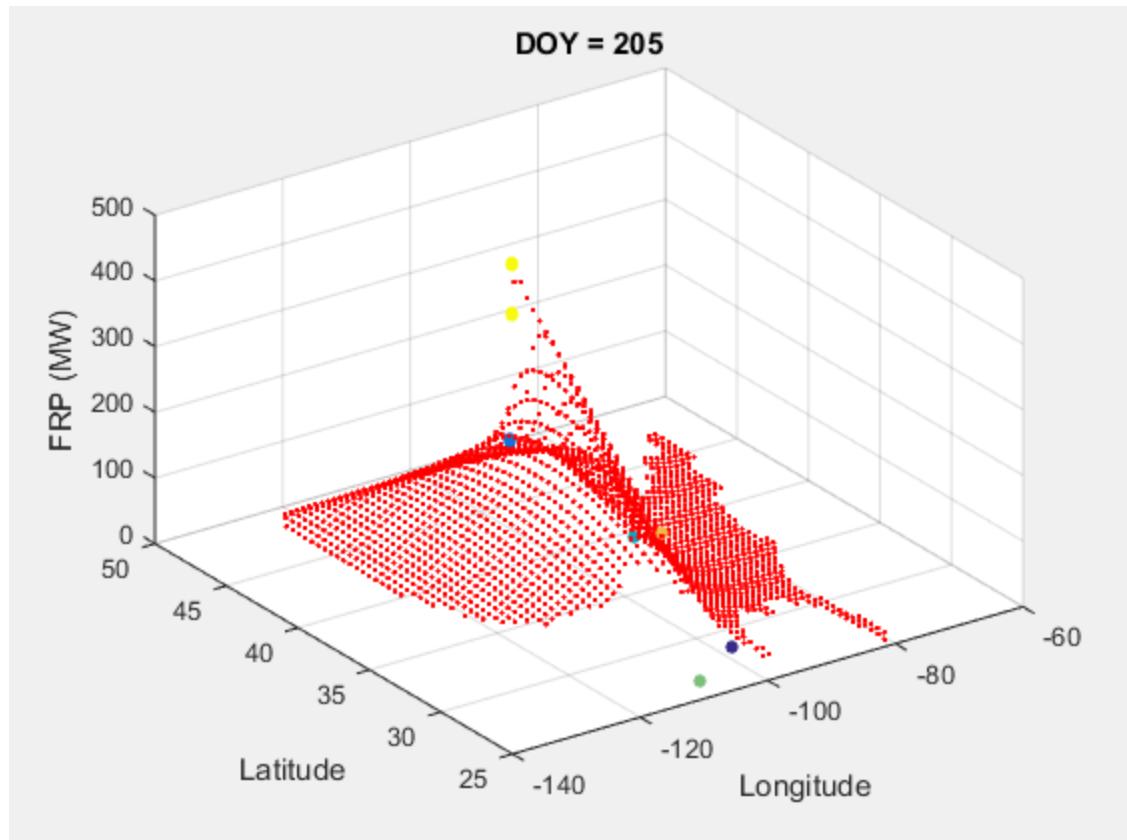


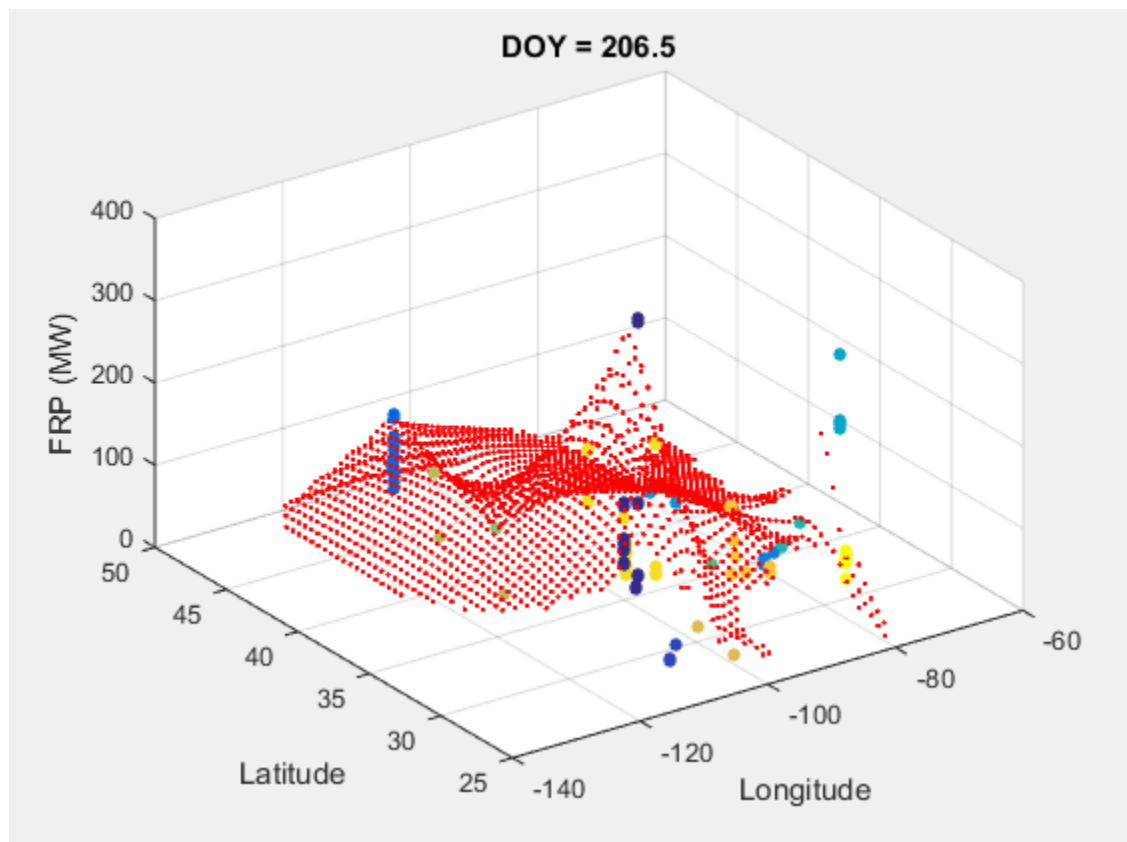
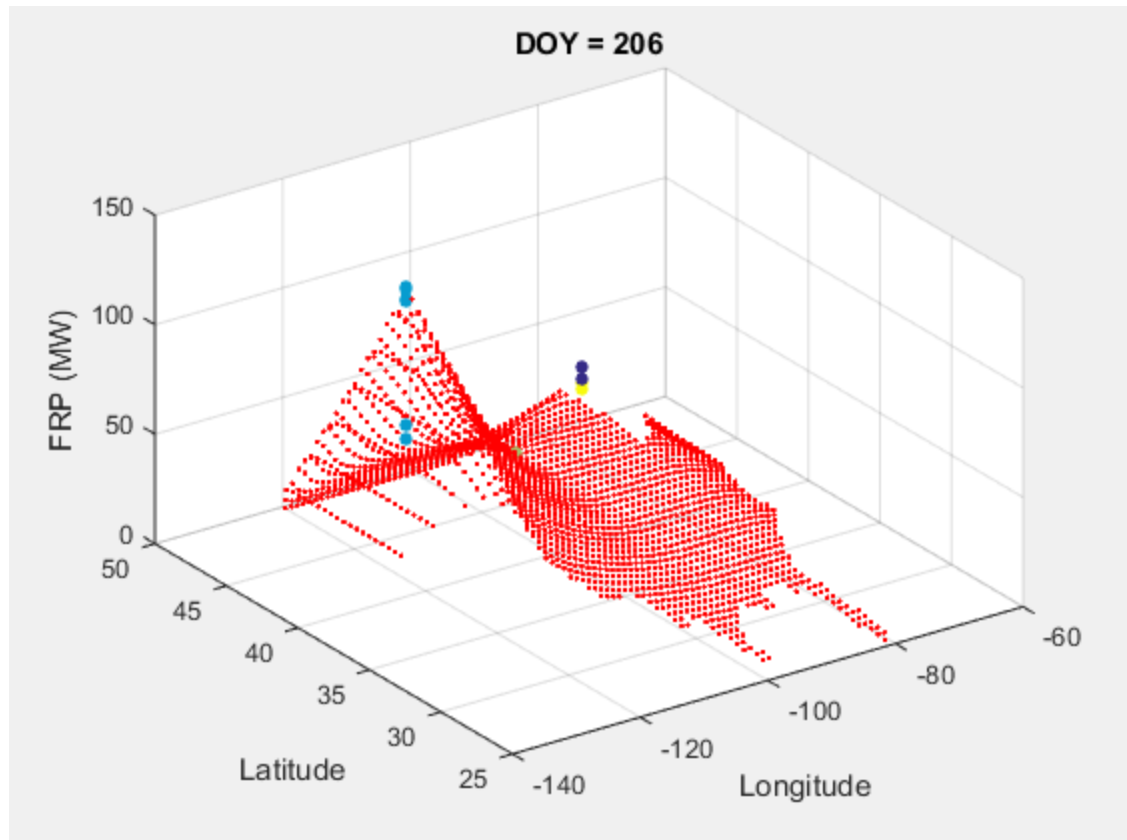


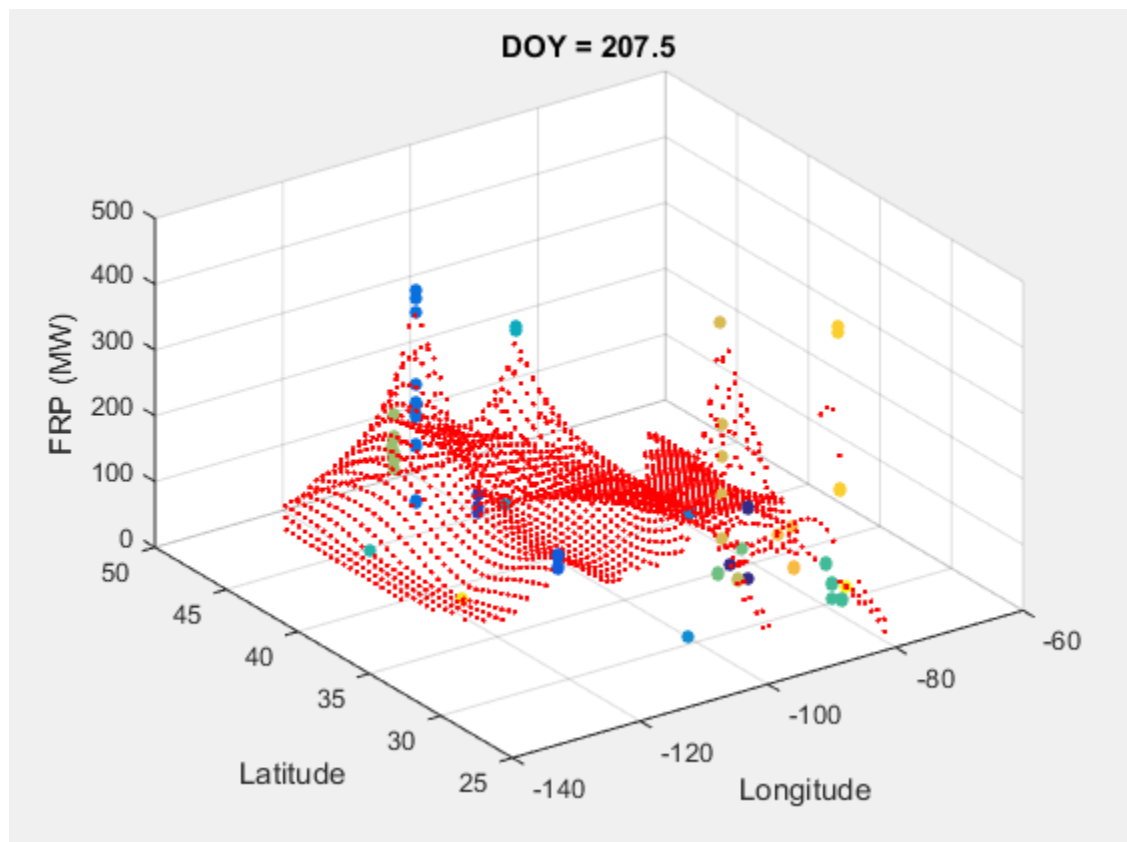
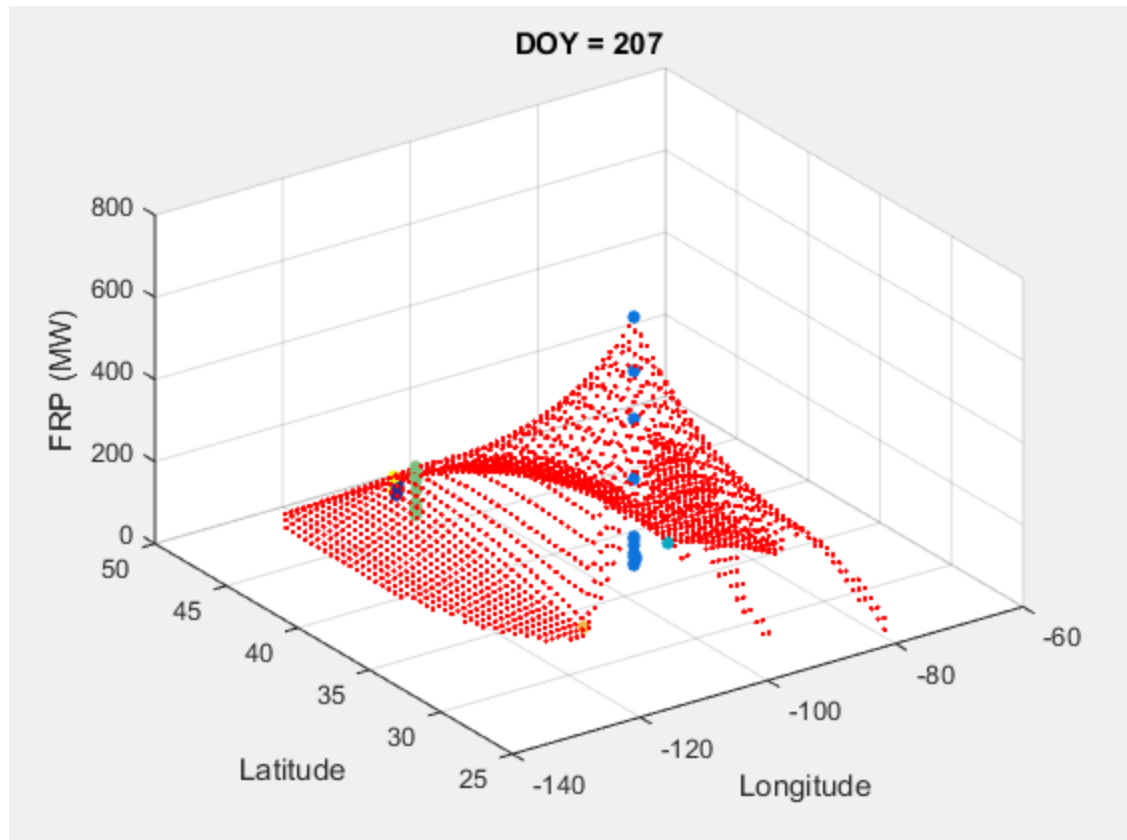


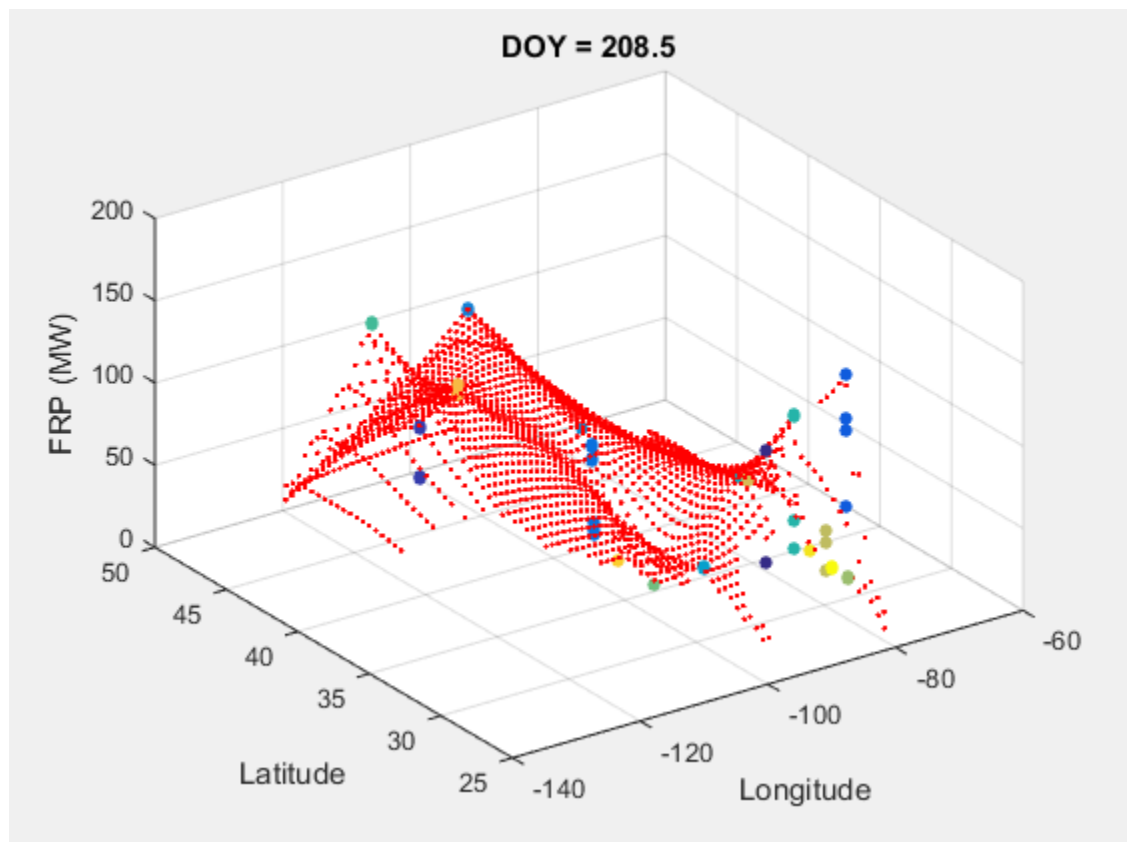
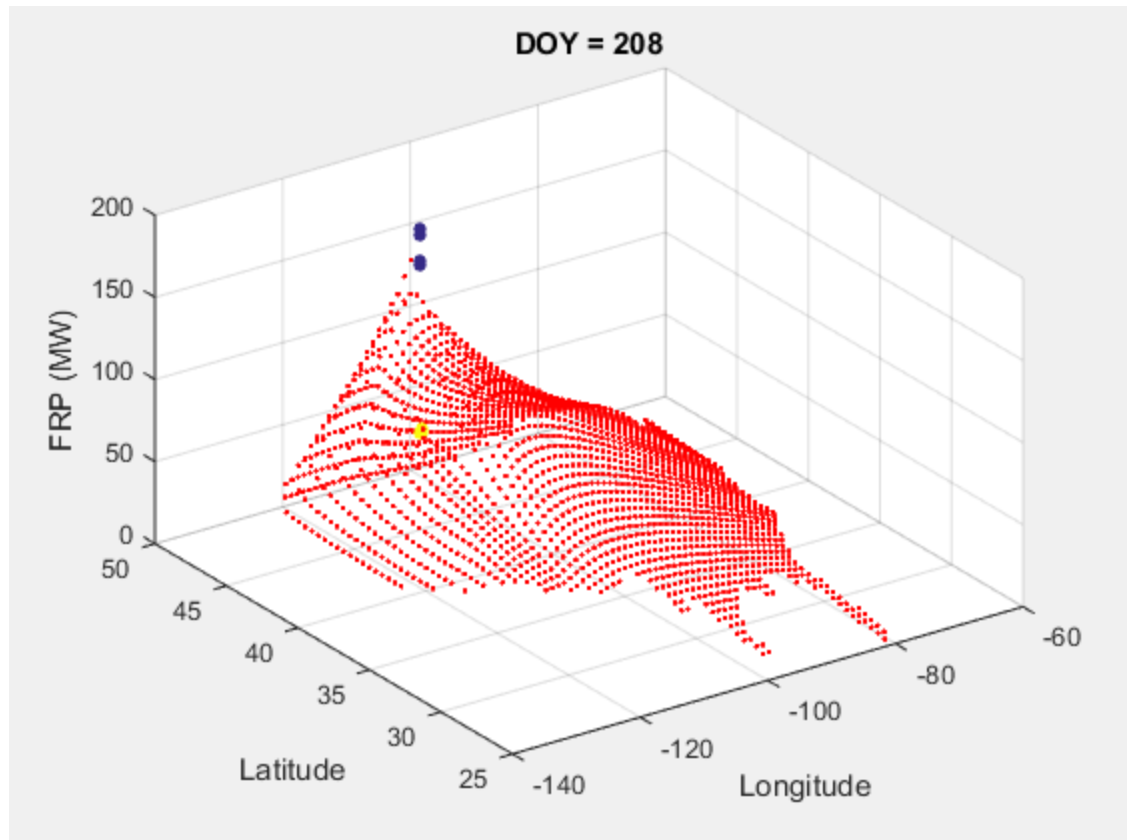


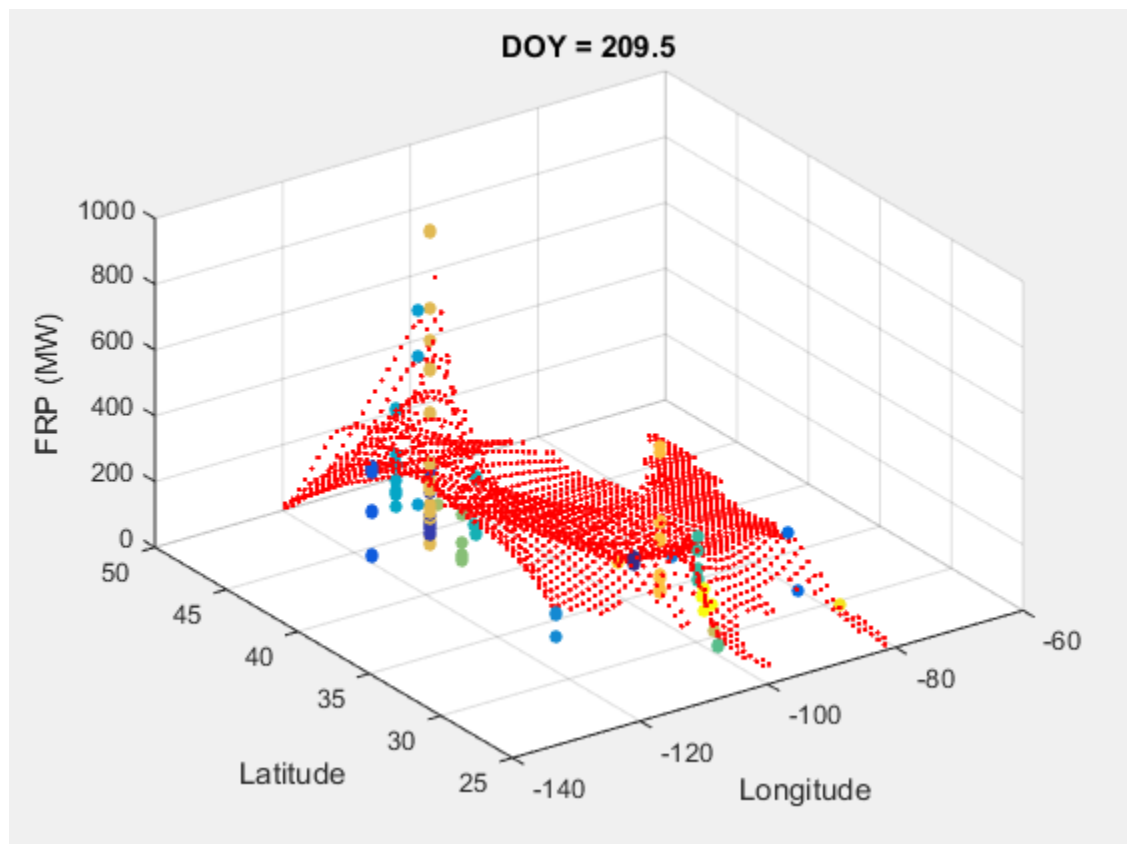
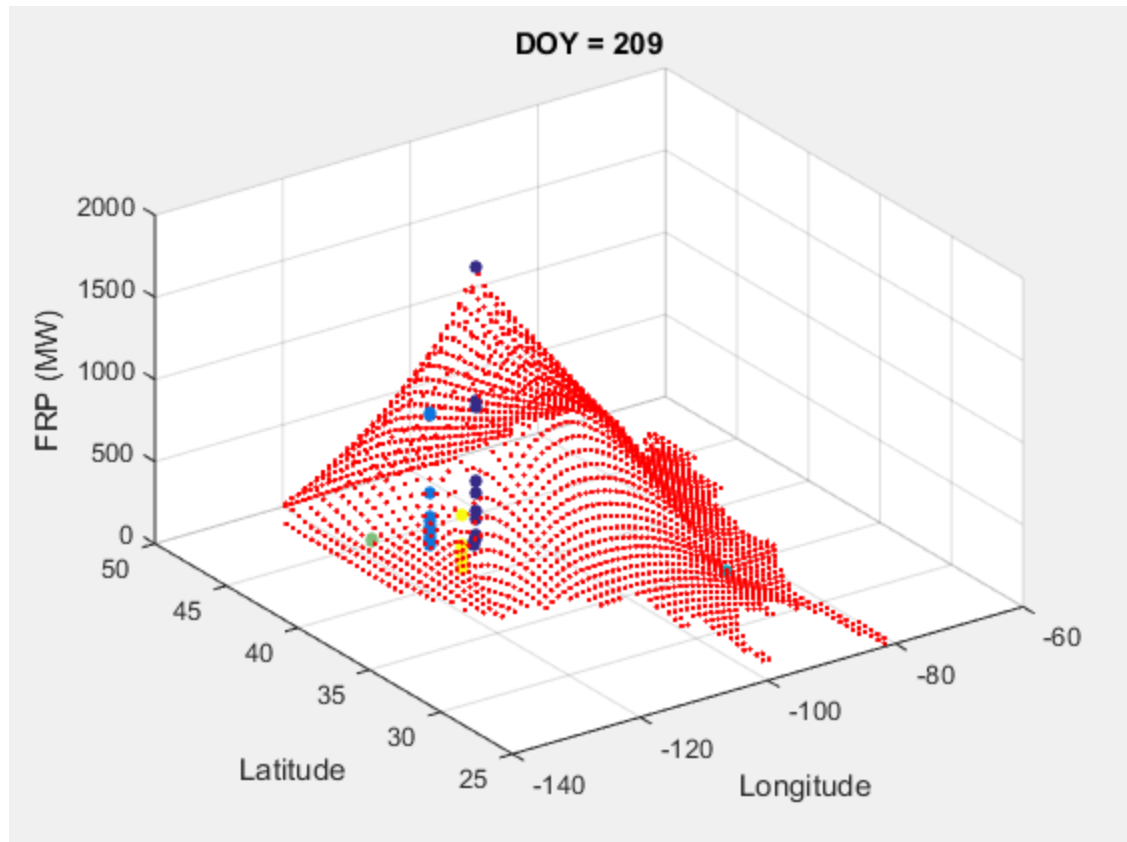












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